

Lecture 6
3.04.2026

The exponential matrix and linear systems
with c.c.

Recall: for a matrix $A \in M_2(\mathbb{R})$, $e^A = \exp(A) = \sum_{k=0}^{\infty} \frac{1}{k!} A^k$

$$e \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \begin{pmatrix} e^{\lambda_1} & 0 \\ 0 & e^{\lambda_2} \end{pmatrix}$$

Ex: Compute $e \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix}$

→ diagonal: if the elements outside the main diagonal are 0.
→ this one is NOT diagonal.

$$A = \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix} = \beta \cdot \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$I_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, I_2^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$A = \beta \cdot I_2$$

We need the powers of I_2 .

$$I_2^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I_2.$$

$$I_2^3 = -I_2 \cdot I_2 = -I_2$$

$$I_2^4 = (-I_2) \cdot I_2 = -I_2^2 = I_2$$

$$I_2^{2k} = (-1)^k \cdot I_2 \quad \left. \begin{array}{l} \\ \end{array} \right\} , (\forall) k \in \mathbb{N}.$$

$$I_2^{2k+1} = (-1)^k \cdot I_2$$

$$\begin{aligned} e \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix} &= e^{\beta \cdot I_2} = \sum_{n=0}^{\infty} \frac{\beta^n}{n!} I_2^n = \\ &= \sum_{\substack{n=0 \\ \text{even}}}^{\infty} \frac{\beta^n}{n!} I_2^n + \sum_{\substack{n=0 \\ \text{odd}}}^{\infty} \frac{\beta^n}{n!} I_2^n = \sum_{k=0}^{\infty} \frac{\beta^{2k}}{(2k)!} I_2^{2k} + \sum_{k=0}^{\infty} \frac{\beta^{2k+1}}{(2k+1)!} I_2^{2k+1} \end{aligned}$$

- exp. matrix
- diag. matrix
- eigenval., eigen vect.
- $x' = Ax$ using diag...

$$= \underbrace{\left(\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} \cdot \beta^{2k} \right)}_{\cos \beta} I_2 + \underbrace{\sum_{k=0}^{\infty} \frac{(-1)^k \cdot \beta^{2k+1}}{(2k+1)!}}_{\sin \beta} I_2$$

$$= (\cos \beta) I_2 + (\sin \beta) I_2$$

$$\Rightarrow e^{\begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix}} = \begin{pmatrix} \cos \beta & \sin \beta \\ -\sin \beta & \cos \beta \end{pmatrix}$$

The exponential of a matrix $A \in \mathcal{M}_m(\mathbb{C})$:

$$e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k \quad (\text{it is also abs. convergent})$$

Ex: $e^{a_m} = I_m$

- if $\exists p \in \mathbb{N}^*$ s.t. $A^p = O_2$ then $A^k = O_2, (\forall) k \geq p$

$$\Rightarrow e^A = I_2 + A + \frac{1}{2!} A^2 + \dots + \frac{1}{(p-1)!} A^{p-1}$$

- if $A = \text{diag}(\lambda_1, \dots, \lambda_m) = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_m \end{pmatrix}$ then

$$A^k = \text{diag}(\lambda_1^k, \dots, \lambda_m^k), (\forall) k \in \mathbb{N}, \text{ and}$$

$$e^A = \text{diag} \left(\sum_{k=0}^{\infty} \frac{1}{k!} \lambda_1^k, \dots, \sum_{k=0}^{\infty} \frac{1}{k!} \lambda_m^k \right) = \text{diag}(e^{\lambda_1}, \dots, e^{\lambda_m}), (\forall) \lambda_1, \dots, \lambda_m \in \mathbb{C}$$

Some Linear Algebra.

- Let $A \in \mathcal{M}_m(\mathbb{C})$. An eigenvalue of A is a number $\lambda \in \mathbb{C}$ s.t.

$\exists$ a vector, called eigenvector, $u \in \mathbb{C}^m \setminus \{0\}$ s.t. $Au = \lambda u$

$$(\Leftrightarrow) u \in \text{Ker}(A - \lambda I_m)$$

- We say that the matrix $A \in \mathcal{M}_m(\mathbb{C})$ is similar to the matrix $B \in \mathcal{M}_m(\mathbb{C})$ when $\exists P \in \mathcal{M}_m(\mathbb{C})$ invertible s.t. $A = P \cdot B \cdot P^{-1}$ ($\Leftrightarrow$) $AP = PB$

transition
matrix

- Property: Two similar matrices have the same eigenvalues.

Proof:

- We have $A = PBP^{-1}$.
- Let $\lambda \in \mathbb{C}$ be an eigenvalue of A .
- Then $\exists u$ an eigenvector, i.e. $u \in \mathbb{C}^m \setminus \{0\}$ and $Au = \lambda u$.

$$PBP^{-1} \cdot u = \lambda u \quad | \cdot P^{-1}$$

$$B(P^{-1} \cdot u) = \lambda (P^{-1} \cdot u)$$

by def.

$\Rightarrow \lambda$ is an eigenvalue of B with the eigenvector $P^{-1}u$.

Remarks.

1) A is similar to B with the transition matrix $P \Rightarrow$

$\Rightarrow B$ is similar to A with the trans. matrix P^{-1} .

2) $\det(A - \lambda I_m)$ is a polynomial in λ of degree m .

$\hookrightarrow$ CHARACTERISTIC POLYNOMIAL of A . The set of all its roots is

denoted $\delta(A) = \{\lambda_1, \dots, \lambda_m\} \subset \mathbb{C}$

spectrum of A .

Lemma.

If $A = PBP^{-1}$, then $A^k = PB^kP^{-1}$, $k \in \mathbb{N}$ and $e^A = P \cdot e^B \cdot P^{-1}$

Proof: We use induction to prove the first relation.

$$k=0 \Rightarrow I_2 = I_2$$

$k=1 \Rightarrow$ hypothesis.

• We assume that, for a fixed k , we have $A^k = P \cdot B^k \cdot P^{-1}$

and we prove that $A^{k+1} = P \cdot B^{k+1} \cdot P^{-1}$

$$A^{k+1} = A^k \cdot A = (P \cdot B^k \cdot P^{-1}) \cdot A = (P \cdot B^k \cdot P^{-1}) \cdot \underline{A} =$$

$$= P \cdot B^k \cdot P^{-1} \cdot \underbrace{P \cdot B \cdot P^{-1}}_{I_m} = P \cdot \underbrace{B^k \cdot B}_{B^{k+1}} \cdot P^{-1} = P \cdot B^{k+1} \cdot P^{-1}$$

Second relation:
$$e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k = \sum_{k=0}^{\infty} \frac{1}{k!} \cdot P \cdot B^k \cdot P^{-1}$$

$$= P \left(\sum_{k=0}^{\infty} \frac{1}{k!} B^k \right) P^{-1} = P \cdot e^B \cdot P^{-1}$$

Def: Given a matrix $A \in M_n(\mathbb{C})$, its Jordan canonical form is a matrix $B \in M_n(\mathbb{C})$ s.t. A is similar to B and B is "the simplest" matrix with this property

Proposition: Let $A \in M_n(\mathbb{C})$ and $\lambda_1, \dots, \lambda_m \in \mathbb{C}$ be the eigenvalues of A , (not necessarily distinct, they can repeat.) Assume that there exist m linearly independent eigenvectors of A . $u_1, \dots, u_m \in \mathbb{C}^n$ (with $Au_k = \lambda_k u_k$).

Let $D = \text{diag}(\lambda_1, \dots, \lambda_m)$ and $P = (u_1, \dots, u_m)$. Then P is invertible. Then $A = PDP^{-1}$ i.e. A is similar to a diagonal matrix. and

Proof.

$$A = PDP^{-1} \quad \Rightarrow \quad AP = PD$$

$$AP = A(u_1, \dots, u_m) = (Au_1 \quad Au_2 \quad \dots \quad Au_m) = (\lambda_1 u_1 \quad \lambda_2 u_2 \quad \dots \quad \lambda_m u_m).$$

$\underbrace{\hspace{1cm}}$ the first column of P $\underbrace{\hspace{1cm}}$ the first column

$$PD = (u_1 \quad \dots \quad u_m) \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_m \end{pmatrix} = (\lambda_1 u_1 \quad \lambda_2 u_2 \quad \dots \quad \lambda_m u_m)$$

$$\Rightarrow AP = PD$$

In this case, D is the Jordan canonical form of A .

Why lin. indep.? Then P is invertible....

Def: We say that A is diagonalizable over $\mathbb{C}$ when A is similar to a diagonal matrix.

Proposition: A is diagonalizable over $\mathbb{C}$ if and only if it has n lin. indep. eigenvectors in $\mathbb{C}^n$.

Proof:

We need to prove $\Rightarrow$. Assume that A is diagonalizable over $\mathbb{C}$.

$\Rightarrow A = PDP^{-1}$ with D a diagonal matrix and P invertible.

D is diagonal $\Rightarrow$ the eigenvalues of D are the numbers of the main diagonal of D .

• A and D are similar $\Rightarrow$ the eigenvalues of D are the eigenvalues of A .

• Denote by $\lambda_1, \dots, \lambda_m$ the eigenvalues of A .

$\Rightarrow$ Then $D = \text{diag}(\lambda_1, \dots, \lambda_m)$

• Denote by $u_1, \dots, u_m \in \mathbb{C}^m$ the columns of P . Since P is invertible, we have $u_1, \dots, u_m$ are linearly independent (1)

$$A = PDP^{-1} \Leftrightarrow AP = PD.$$

$$\left. \begin{aligned} AP &= (A u_1 \dots A u_m) \\ PD &= (\lambda_1 u_1 \dots \lambda_m u_m) \end{aligned} \right\} \Rightarrow A u_k = \lambda_k u_k. \quad (2)$$

(1), (2) $\Rightarrow u_1, \dots, u_m$ are m lin. indep. eigenvectors of A .

Def: A matrix which is not diagonalizable over $\mathbb{C}$ is called defective.

Example of defective matrix:

$$A = \begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}, \quad a \in \mathbb{C}.$$

The eigenvalues of A are $\lambda_1 = a, \lambda_2 = a$ (double)

$$u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

$$Au = au \Leftrightarrow (A - a \cdot I_2)u = 0$$

$$(2) \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = 0$$

$$(3) \quad u_2 = 0$$

$$u = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \text{this is the only eigenvector.}$$

$\Rightarrow$ There are now 2 lin. indep. eigenvectors for A . Thus, A is defective.

Prop: If any ~~no~~ eigenvalue of A is simple, then A is diagonalizable over $\mathbb{C}$. ~~Ex: A~~

Example: $A = \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix}$, $\beta \in \mathbb{R}^*$

$$\det(A - \lambda I_2) = \begin{vmatrix} -\lambda & \beta \\ -\beta & -\lambda \end{vmatrix} = \lambda^2 + \beta^2 = 0 \Rightarrow \lambda^2 = -\beta^2$$

$$\Rightarrow \lambda = \pm i\beta$$

$$\boxed{\lambda^2 - (\text{tr} A)\lambda - \det A = 0}$$

$$\text{tr} A = 0$$

$$\det A = \beta^2$$

$$\lambda^2 + \beta^2 = 0$$

$$\lambda^2 = -\beta^2$$

$$\lambda_{1,2} = \pm i\beta \rightarrow \begin{cases} \lambda_1 = i\beta \\ \lambda_2 = -i\beta \end{cases}$$

$$\lambda_1 \neq \lambda_2 \Rightarrow \text{simple}$$

(they don't repeat)

$\Rightarrow A$ diagonalizable over $\mathbb{C}$.

Let $u_1 \in \mathbb{C}^2 \setminus \{0\}$ be s.t. $Au_1 = i\beta u_1$, then $u_2 = \bar{u}_1$ (the complex conjugate) is the other eigenvector, $Au_2 = (-i\beta)u_2$

$$P = (u_1, u_2) \Rightarrow A = P \text{diag}(i\beta, -i\beta)$$

$$A = P \cdot \begin{pmatrix} i\beta & 0 \\ 0 & -i\beta \end{pmatrix} \cdot P^{-1}$$

$$\Rightarrow e^A = P \cdot \begin{pmatrix} e^{i\beta} & 0 \\ 0 & e^{-i\beta} \end{pmatrix} \cdot P^{-1}$$

Def: $A \in \mathcal{M}_m(\mathbb{R})$ is diagonalizable over $\mathbb{R}$ when A is similar to a real diagonal matrix.

Remark! Let $A \in \mathcal{M}_m(\mathbb{R})$ which is diagonalizable over $\mathbb{C}$. We have that A is diag. over $\mathbb{R}$ iff $\sigma(A) \subset \mathbb{R}$.

Linear diff. systems with constant coeff.

Theorem 1. Let $A \in \mathcal{M}_m(\mathbb{R})$, and $\eta \in \mathbb{R}^m$. Then, IVP $x' = Ax$, $x(0) = \eta$ has a unique sol., given by $\varphi(t, \eta) = e^{tA} \cdot \eta$, $t \in \mathbb{R}$.

Remark: The general sol. of $x' = Ax$ is $x(t) = e^{tA} \cdot c$, where $c \in \mathbb{R}^m$ is arbitrary. (vector of c.) $c = \begin{pmatrix} c_1 \\ \vdots \\ c_m \end{pmatrix}$

The general sol. of $x' = AX$ when A is diagonalizable over $\mathbb{C}$.

Step 1: Find the eigenvalues $\lambda_1, \dots, \lambda_m \in \mathbb{C}$ of A .

Step 2: Find the m lin. indep. eigenvectors $u_1, \dots, u_m$ of A

Step 3: $D = \text{diag}(\lambda_1, \dots, \lambda_m)$

$P = (u_1, \dots, u_m)$

Step 4: The general sol:

$$x(t) = P \text{diag}(e^{t\lambda_1}, \dots, e^{t\lambda_m}) P^{-1} \cdot c, \quad c \in \mathbb{R}, \text{ arbitrary}$$

$$A = PDP^{-1} \Rightarrow e^{tA} = P \cdot e^{tD} \cdot P^{-1}$$